BLEKINGE TEKNISKA HÖGSKOLA

BTH

Gender-Based-Comparative-Analysis-of-Heart-Disease-Prediction-Models

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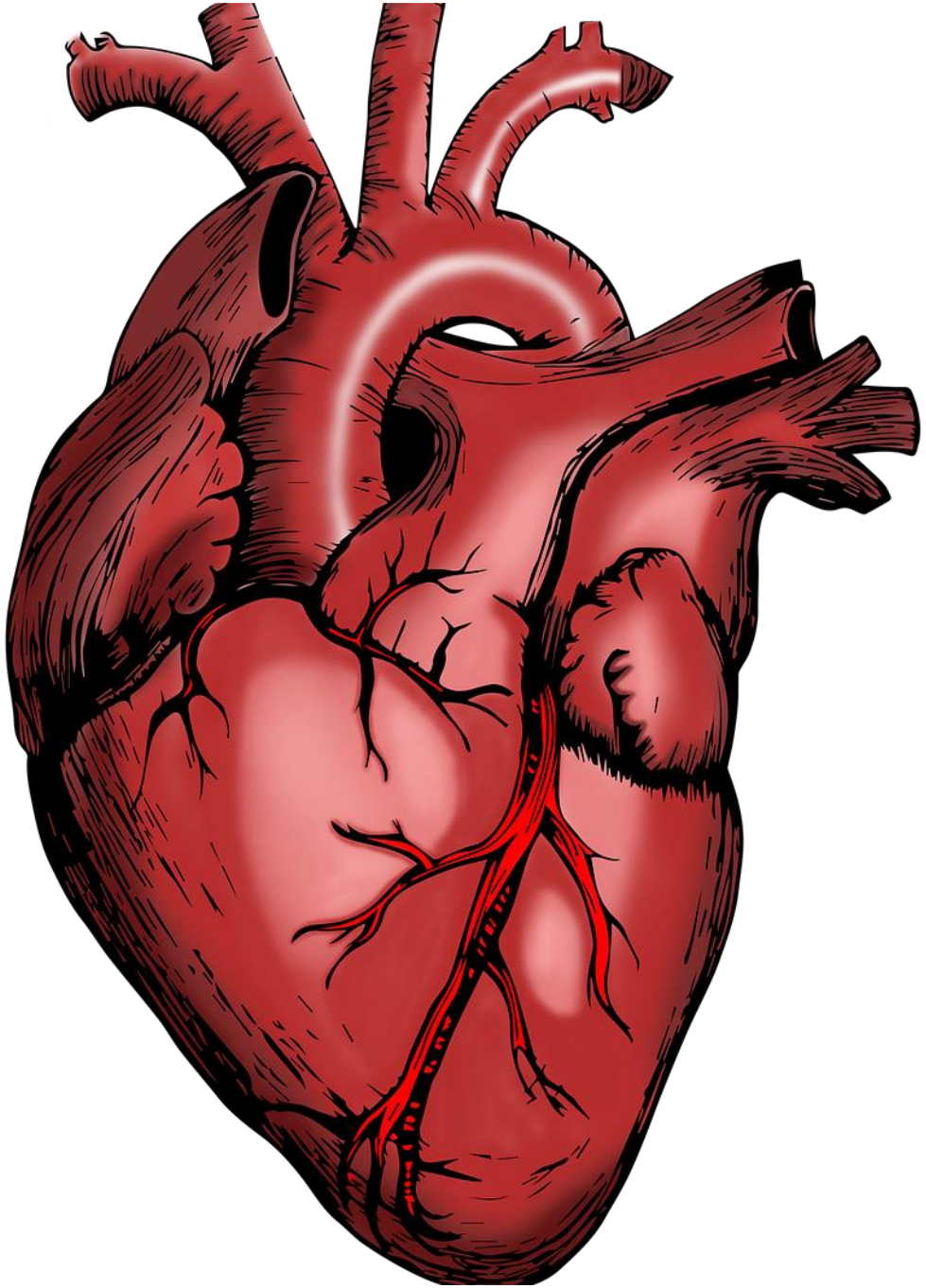
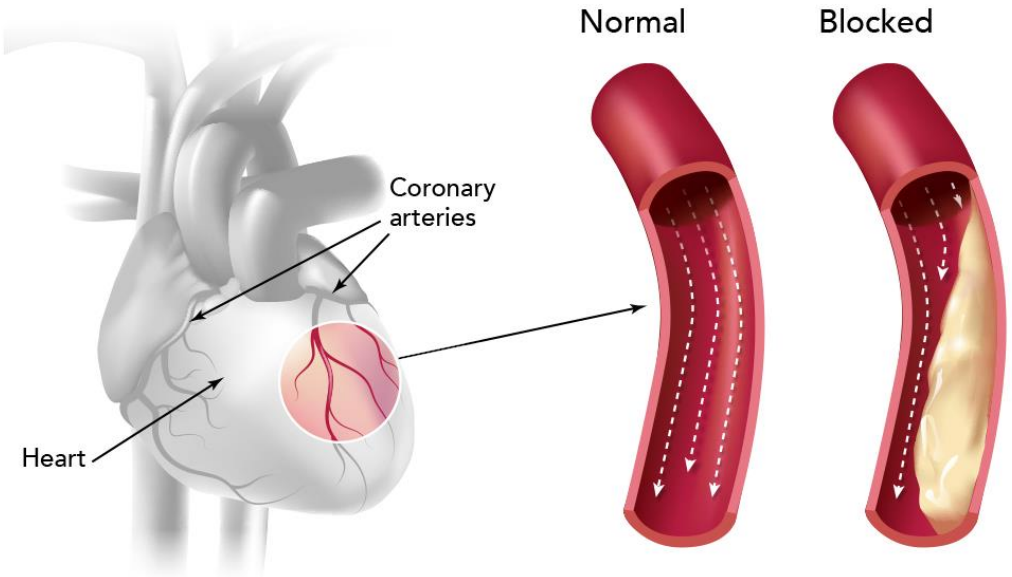


Background

- Cardiovascular Disease - heart's arteries narrow or become blocked.
- 19.8 million deaths in 2022 - 32% of all global deaths [1].
- Timely prediction - preventive interventions.



[1] https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)

Problem Statement

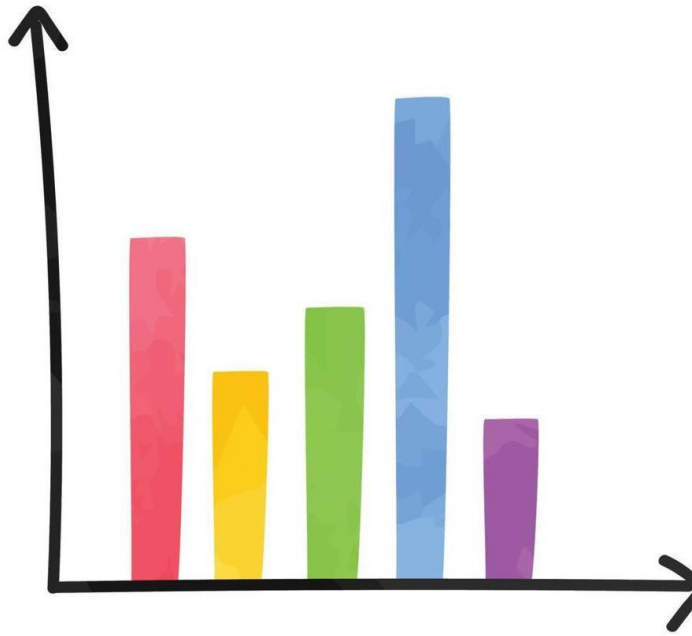
- Develop a system for predicting the presence of heart disease based on clinical parameters.
- Also, investigate gender-specific disease manifestation patterns.

Solution

- Train and compare 3 ML models:
  - Logistic Regression (Baseline)
  - Random Forest
  - XGBoost
  - Ensemble

Dataset

- Heart Disease Dataset, Kaggle
- Size: 1,025
- Features: 13 clinical attributes
- Target: Binary classification
- Numerical & Categorical features – original source



Performance Metrics

- Accuracy: Overall correctness
- Precision: Positive predictive value
- **Recall** (Sensitivity): **minimizing false negatives**
- F1-Score
- ROC
- Confusion Matrix

Questions & Discussion

Thank You